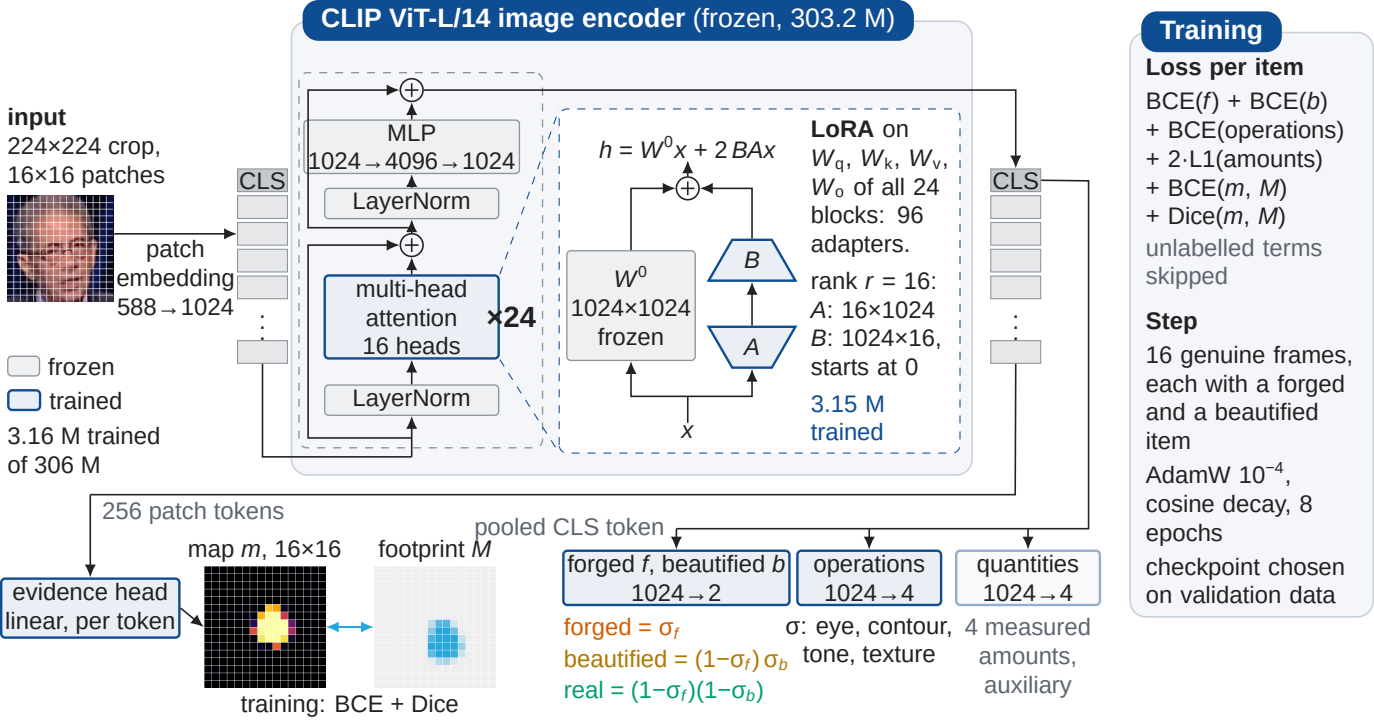

CLIP ViT-L/14 image encoder (frozen, 303.2 M)
input
224×224 crop,
16×16 patches
patch
embedding
588 → 1024
CLS
MLP
1024 → 4096 → 1024
LayerNorm
multi-head
attention
16 heads
×24
LayerNorm
h = W⁰x + 2 BAx
W⁰
1024×1024
frozen
B
A
x
LoRA on
W_q, W_k, W_v,
W_o of all 24
blocks: 96
adapters.
rank r = 16:
A: 16×1024
B: 1024×16,
starts at 0
3.15 M
trained
CLS
frozen
trained
3.16 M trained
of 306 M
256 patch tokens
pooled CLS token
map m, 16×16
footprint M
evidence head
linear, per token
training: BCE + Dice
forged f, beautified b
1024 → 2
operations
1024 → 4
quantities
1024 → 4
forged = σ_f
beautified = (1−σ_f) σ_b
real = (1−σ_f)(1−σ_b)
σ: eye, contour,
tone, texture
4 measured
amounts,
auxiliary
Training
Loss per item
BCE(f) + BCE(b)
+ BCE(operations)
+ 2·L1(amounts)
+ BCE(m, M)
+ Dice(m, M)
unlabelled terms
skipped
Step
16 genuine frames,
each with a forged
and a beautified
item
AdamW 10⁻⁴,
cosine decay, 8
epochs
checkpoint chosen
on validation data
page: MediaPipe landmarks → part masks → named part
nose (mean evidence 0.99 on this held-out forgery)